✅ 🗓️ DAY 2 — EMBEDDINGS + VECTOR RAG (UPDATED FOR YOU)

🎯 Goal

👉 Build **REAL RAG using local embedding (Qwen)**

✅ 🔧 CHANGE (IMPORTANT — YOUR VERSION)

Instead of HuggingFace ❌ 👉 You used:

ollama.embeddings(model="qwen2.5") ✅ This is perfect for your restriction

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✅ FINAL FILES (IMPORTANT)

📁 embeddings/embedding_service.py

✅ YOUR VERSION:

```
import ollama

class EmbeddingService:

  def encode(self, texts):

    if isinstance(texts, str):

      texts = [texts]

    embeddings = []

    for text in texts:

      response = ollama.embeddings(

        model="qwen2.5", prompt=text )

      embeddings.append(response["embedding"])

    return embeddings
```

📁 **embeddings/vector_store.py**

✅ No change

📁 **retrieval/vector_search.py**

✅ FINAL:

```
from embeddings.embedding_service import EmbeddingService
from embeddings.vector_store import VectorStore
class VectorSearch:
  def init(self):
    self.embedder = EmbeddingService()
    self.store = VectorStore()
  def index(self, documents):
    embeddings = self.embedder.encode(documents)
    ids = [str(i) for i in range(len(documents))]
    metadata = [{"source": "file"} for _ in documents]
    self.store.add(ids, documents, embeddings, metadata)
  def search(self, query):
    query_embedding = self.embedder.encode(query)[0]
    results = self.store.query(query_embedding)
    return results
```

📁 **app/main.py (UPDATED)**

```
from retrieval.vector_search import VectorSearch
from ingestion.loader import load_text_file
from models.llm import generate
```

```
def main():
    print("Multimodal AI System (RAG Enabled)")
    vs = VectorSearch()
    filepath = "data/raw/sample.txt"
    docs = load_text_file(filepath)
    vs.index(docs)
    print("Documents indexed!")
    while True:
        query = input(">> ")
        if query.lower() == "exit":
            break
        results = vs.search(query)
        context = "\n".join(results)
        prompt = f"""
Answer only from the context below.
Context:
{context}
Question:
{query}
"""
        response = generate(prompt)
        print("\n", response)
if __name__ == "__main__":
    main()
```

✅ **OUTPUT NOW**

Documents indexed!

>> what is ai

✅ Retrieval working
✅ Context being used
✅ Qwen answering

🧠 **WHAT YOU HAVE BUILT (IMPORTANT)**

👉 You now have:

Component	Status
LLM	✅ Qwen
Embedding	✅ Qwen embeddings
Vector DB	✅ Chroma
Retrieval	✅ Semantic search
Pipeline	✅ End-to-end

👉 This is:

✅ ✅ **REAL WORKING RAG SYSTEM (LOCAL)**

⚠️ **LIMITATION (HONEST)**

Using Qwen embeddings:

- ✅ works
- ❌ not best quality

👉 But perfect for your environment ✅

🚀 WHAT CHANGES FROM ORIGINAL PLAN

❌ Removed:

- HuggingFace embeddings
- internet dependency

✅ Added:

- Ollama embeddings
- Fully local pipeline

✅ ✅ CURRENT LEVEL (IMPORTANT)

You are now at:

LLM user ❌

Basic coder ❌

✅ RAG SYSTEM BUILDER ✅

🚀 NEXT STEP (DAY 3)

Now things get **serious improvement**

👉 We will add:

✅ BM25 (keyword search)
✅ Hybrid retrieval (vector + keyword)
✅ RRF fusion (your original idea 🔥)
